Euler totient function ϕ -

$$\phi(n) = \text{number of integers } x \quad 1 \le x \le n, \ \gcd(x,n)=1$$

$$= |U(n)|$$

ϕ has a multiplicative formula.

If n has prime factorization

$$n = \prod_{i=1}^{r} p_i^{a_i}, \quad p_i \text{ distinct primes}, \quad a_i \geq 1$$

then $$\phi(n) = \prod_{i=1}^{r} p_i^{a_i - 1}(p_i - 1)$$

The conjectures from the previous lecture.

Fermat's Little Theorem

Concerns a prime modulus p.

For any integer a, not divisible by p $(p \nmid a)$, i.e. $a \in U(p)$.

<u>then</u> $a^{p-1} \equiv 1 \pmod{p}$.

(this will imply that $|a|$, order of a, is a factor of $p-1$).

<u>Proof</u> (number theoretic).

Consider the product of all non-zero congruence classes in two ways.

- Consider $\prod_{j=1}^{p-1} j = (p-1)!$
- Consider the multiples of a congruence classes modulo p

$$a, 2a, 3a, \ldots, (p-1)a \quad (*)$$

- <u>Claim:</u> These are a permutation of $1, 2, 3, \ldots, p-1$

Pf: Suppose $ja \equiv ka \pmod{p}$

$\Rightarrow ja a^{-1} \equiv k a a^{-1}$, a^{-1} exists since $p \nmid a$. $\gcd(a,p)=1$

$\Rightarrow j \equiv k$

Therefore the elements in $(*)$ are distinct modulo p. and so are <u>all</u> the non-zero congruence classes

modulo p.

Therefore $\prod_{j=1}^{p-1} ja \equiv (p-1)! \pmod{p}$

$$\Longrightarrow a^{p-1} \prod_{j=1}^{p-1} j \equiv (p-1)!$$

$$\Longrightarrow a^{p-1} (p-1)! \equiv (p-1)!$$

$(p-1)!^{-1}$ exists since $\gcd((p-1)!, p) = 1$

$$\Longrightarrow a^{p-1} \equiv 1 \pmod{p}.$$

as required.

A corollary of this is that $|a|$ divides $p-1$.

ie. $|\langle a \rangle|$ divides $|U(p)|$

a special case of Lagrange's Theorem.

For a general modulus n, the corresponding result is

<u>Euler's Theorem</u>

For a positive integer n and an integer a such that $\gcd(a,n)=1$ (ie. $a \in U(n)$).

$$a^{\phi(n)} \equiv 1 \pmod{n}.$$

<u>Proof</u>: Repeat the proof idea from Fermat's Little Theorem, but instead of $(p-1)! = \prod_{j=1}^{p-1} j$

we consider

$$\prod_{j \in U(n)} j$$

and remembering $|U(n)| = \phi(n)$ ∎

Corollary.

For any $x \in U(n)$

$|\langle x \rangle|$ divides $\phi(n) = |U(n)|$

Application of Euler's theorem, (and orders of elements, to calculating large powers modulo n.

Consider

$$X^M \equiv ? \pmod{n}$$

for large M, such that the raw integer X^M is inaccessible

Eulers theorem says provided $\gcd(X,n)=1$

$$X^{\phi(n)} \equiv 1 \pmod{n}$$

Use this to simplify M.

Consider integer division of M by $\phi(n)$.

$$M = q\phi(n) + m, \text{ for some integers } q, m \quad 0 \le m < \phi(n)$$

large exponent -

$$X^M = X^{q\phi(n)+m}$$

$$= \left(X^{\phi(n)}\right)^q X^m, \text{ exponent rules.}$$

$$\equiv 1^q \cdot x^{-m} \pmod{n}$$

, by Euler's theorem

$$\equiv x^{m} \pmod{n}.$$

small exponent

$0 \le m < \phi(n)$

Can also reduce the base modulo n if this helps using.

If $a \equiv b \pmod{n}$

then $a^e \equiv b^e \pmod{n}$

All of this relevant for Q2.

Example 5.1 More applications of <u>multiplicative formula</u> for ϕ.

Continue at 4:55.

Q2. Find all the integers n such that $\phi(n) = 12$.

When B, when $n = \prod_{i=1}^{r} p_i^{a_i}$

$$\phi(n) = \prod_{i=1}^{r} p_i^{a_i - 1}(p_i - 1) = 12 \; ?$$
$$= 2^2 \cdot 3$$

This surely places great restrictions on the primes p_i that might occur here

- No prime $p > 13$ can occur, since this would contribute a factor > 12 to $\phi(n)$.

- So the possibilities.

$$p = 2, 3, 5, 7, \cancel{11}, 13$$

note if $p = 11$, $p - 1 = 10$ which does not divide 12.

So now we know | any solution n of $\phi(n) = 12$ must take the form

$$n = 2^a 3^b 5^c 7^d 13^e, \text{ for}$$

some exponents $0 \leq a, b, c, d, e$

But surely there are restrictions

on exponents.

$e = 0, 1$, since $13 \nmid 12$. $\phi(n) = 12$
$= 2^2 \cdot 3$

$d = 0, 1$, since $7 \nmid 12$

$c = 0, 1$, since $5 \nmid 12$.

$b = 0, 1, 2.$, since $3^2 \nmid 12$.

$a = 0, 1, 2, 3.$, since $2^3 \nmid 12$.

Now we know there are at most 96 solutions n.

Proceed on a cases by cases basis.

<u>Suppose $e = 1$</u>

ie. $n = m \cdot 13$ $\quad \gcd(m, 13) = 1$

$$\Rightarrow \phi(n) = \phi(m \cdot 13)$$
$$= \phi(m)\phi(13)$$
$$= 12\phi(m)$$
$$= 12.$$

$$\Rightarrow \phi(m) = 1$$

$$\Rightarrow m = 1, 2$$

For any $m > 3$, $\phi(m) \geq 2$, since $1, m-1$ are coprime to m

Therefore $n = 26, 13$ are two solutions to $\phi(n) = 12$.

Now we assume $e = 0$.

Now suppose $\underline{d = 1}$

$\Rightarrow n = 7m$, $\gcd(m, 7) = 1$

$$\Rightarrow \phi(n) = \phi(7m)$$
$$= \phi(7)\phi(m)$$
$$= 6\phi(m) = 12.$$

$\Rightarrow \phi(m) = 2$.

$\Rightarrow \underline{m = 3, 4, 6} \; (= 3^1, 2^2, 2 \cdot 3)$

Therefore $n = 21, 28, 42$ are solutions to $\phi(n) = 12$.

From now on $d = 0$, $e = 0$

Now suppose $c=1$.

$\Rightarrow n = 5m$, $\gcd(m,5)=1$

$\Rightarrow \phi(n) = \phi(5m)$

$\Rightarrow \phi(n) = \phi(5)\phi(m)$

$= 4\phi(m) = 12.$

$\Rightarrow \phi(m) = 3.$

But from mult. form. for ϕ,
$\phi(m)$ is always even or $=1$.
So there are no such m
So there are no solutions with $5 \mid n$.

So now we have $e, d, c = 0$

So $n = 2^a 3^b$ $\quad a = 0,1,2,3$
$\quad b = 0,1,2.$

Of these 12 possibilities
only $n = 2^2 \cdot 3^2$ is a solution.
$= 36$

$\phi(n) = 2 \cdot 3 \cdot 2 = 12.$

So we have the six possible solutions

$n = 26, 13, 36, 21, 28, 42.$

Q1.

Claim: $\phi(n) = \phi(2n)$ iff n is odd.

$\Longleftrightarrow$

Suppose n is odd, i.e. $\gcd(2, n) = 1$

$$\Longrightarrow \phi(2n) = \phi(2)\phi(n)$$
$$= \phi(n)$$

Conversely, for the $\Longrightarrow$ direction, a direct proof seems awkward. So let's proving its contrapositive.

(contrapositive of $A \Longrightarrow B$ is $(\neg B) \Longrightarrow (\neg A)$, a logically equivalent statement).

So we will now prove.

If n is even then $\phi(n) \neq \phi(2n)$

<u>Proof</u>: Suppose n is even

$\Rightarrow \quad n = 2 \cdot m$, for some m

$= 2^a \cdot u$, where u is odd and $a \geq 1$

$$\boxed{\phi(2n)} = \phi(2 \cdot 2^a u)$$

$$= \phi(2^{a+1} u)$$, u is odd, $\gcd(2^{a+1}, u) = 1$

$$= \phi(2^{a+1})\phi(u)$$, by the multiplicative formula

$$= \boxed{2^a \phi(u)}$$

But $\boxed{\phi(n)} = \phi(2^a u)$

$$= \boxed{2^{a-1}\phi(u)}.$$

$\Rightarrow \quad \phi(n) \neq \phi(2n)$, as required.

in fact we've proved.

For n even $\quad \phi(2n) = 2\phi(n)$.